

Advanced (Habanero)

Q1. What is the absolute value of -5 ?

- (A) -5
- (B) 5
- (C) 0
- (D) 10
- (E) 25

Q2. Which of the following is equal to $|x|$ when $x = -3$?

- (A) -3
- (B) 3
- (C) 0
- (D) 6
- (E) 9

Q3. The distance between 0 and -8 on the number line is $|8| =$

- (A) -8
- (B) 8
- (C) 16
- (D) 0
- (E) 4

Q4. What is $|-2| + |-7|$ equal to?

- (A) -9
- (B) 9
- (C) 2
- (D) 5
- (E) 14

Q5. The absolute value of 5 is $|$

- (A) -5
- (B) 5
- (C) 10
- (D) 0
- (E) 25

Q6. If $x = -4$, then $|x - 2|$ is

- (A) 2
- (B) 6
- (C) -6
- (D) 4
- (E) 8

Q7. If $|3x| = 9$, then x could be

- (A) 3
- (B) -3
- (C) 1
- (D) -1
- (E) 3 or -3

Q8. The equation $|x| = 4$ has solutions $x =$

- (A) 4
- (B) -4
- (C) 0
- (D) 2
- (E) 4 or -4

Open Ended Questions (FRQ)

Q9. Solve for x in $|2x - 5| = 11$ and show your work.

Q10. Tom's house is $|-15|$ miles from the city center. How far is his house from the city center?

Chef's Tips

- Remind students to read the questions carefully
- Have students show all work for free response questions
- Use a calculator only when necessary